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Scintillation Processing with the Software Receiver

A short Introduction

DLR

Autorenangaben

Name	Organization	Contact
Friederike Wendler	Deutsches Zentrum fr Luft- und Raumfahrt e.V. (DLR)	E-Mail: friederike.wendler@dlr.de Phone: +49 8153 28 1645

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Chapter 1

Introduction

[?]

Chapter 2

Signal Model

The raw received signal from $K \in \mathbb{N}$ satellites and $L \in \mathbb{N}$ carrier frequencies can be modeled by

$$y_{\mathsf{RF}}(t) = \sum_{l=1}^{L} \sum_{k=1}^{K} A_{kl}(t)\, c_{kl}(t - \tau_{kl}(t))\, d_{kl}(t - \tau_{kl}(t)) \cos(2\pi f_{\mathsf{c},l} t + \phi_{kl}(t)) + \eta(t) \tag{2.1}$$

where $A(t) \in \mathbb{R}$ denotes the signal amplitude, and $\tau_{kl}(t) \in \mathbb{R}$ denotes the signal delay including the ionospheric delay for the k-th satellite and l-th carrier frequency. $f_{\mathsf{c},l} \in \mathbb{R}$ denotes the carrier frequency while $\phi_{kl}(t) \in \mathbb{R}$ is the ambiguous phase, which includes the Doppler frequency. $\eta(t) \in \mathbb{R}$ is zero mean white Gaussian noise. The signals $c_{kl}(t) \in \mathbb{R}$ and $d_{kl}(t) \in \mathbb{R}$ denote the code and data bit signals modulated on the l-th carrier signal for the k-th satellite. Note, that the respective signals are well separated by their code and frequency, therefore it is sufficient to consider the received signal from the k-th satellite on the l-th carrier frequency independently in the following

$$y_{\mathsf{RF},kl}(t) = A_{kl}(t)\, c_{kl}(t - \tau_{kl}(t))\, d_{kl}(t - \tau_{kl}(t)) \cos(2\pi f_{\mathsf{c},kl} t + \phi_{kl}(t)) + \eta(t)\,. \tag{2.2}$$

2.1 Analogue Front End

In the following we shortly describe the analogue front end of the measurement setup which is illustrated in Figure 2.1 . For the sake of readability we will omit the subscripts $\bullet_k$, $\bullet_l$ where possible in the following.

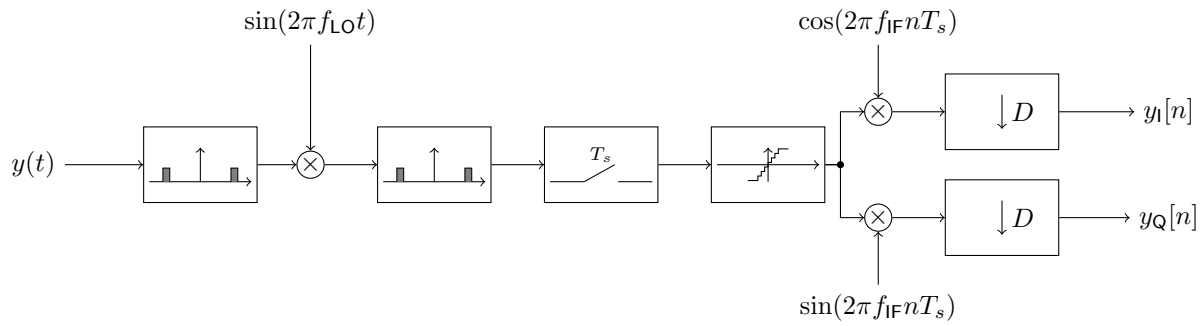


Figure 2.1: Illustration of the Receiver Front-End

2.1.1 Intermediate Frequency Conversion

In a first step the signal is filtered with an analogue bandpass filter to suppress noise and interfering signals outside of the signal band of interest. Next the signal is converted to an intermediate frequency $f_{\mathsf{IF}} \in \mathbb{R}$. This is done by multiplying the signal with a local oscillator tone $f_{\mathsf{LO}} \in \mathbb{R}$ which leads to a signal at intermediate frequency

$$f_{\mathsf{IF}} = f_{\mathsf{c}} - f_{\mathsf{LO}}. \tag{2.3}$$

and an intermodulation product at

$$f_{\mathsf{INT}} = 2f_{\mathsf{c}} - f_{\mathsf{LO}} \tag{2.4}$$

which must be eliminated with a bandpass filter with sufficiently narrow bandwidth. Note, that the local oscillator frequency must not be too small as a signal at the image frequency

$$f_{\mathsf{im}} = f_{\mathsf{c}} - 2f_{\mathsf{IF}} \tag{2.5}$$

would also be converted to intermediate frequency [**?**, p. 134] if not suppressed by a bandpass filter with sufficient selectivity in advance. However, the maximum possible intermediate frequency is determined by the sampling rate of the analogue-to-digital (A/D) converter and the Nyquist criterion. Having an A/D converter sampling rate $f_s \in \mathbb{R}$, f_{IF} has to suffice

$$f_s = f_{\mathsf{IF}} + \frac{B_{c(t)}}{2} \tag{2.6}$$

where $B_{c(t)} \in \mathbb{R}$ is the code signal bandwidth. In the measurement setup with the National Instruments data grabber the front-ends provide $f_{\mathsf{IF}} = 75\,\mathsf{MHz}$. Additionally, the signal is amplified with an automatic gain control (AGC) to have enough power for the A/D converter

2.1.2 A/D Conversion

A/D conversion includes two steps:

1. discrete time sampling
2. discrete value mapping

The signal after A/D conversion is given by

$$y_{\mathsf{A/D}}[n] = Q_B(A[n]\, c(nT_s - \tau[n])\, d(nT_s - \tau[n]) \cos(2\pi f_{\mathsf{IF}} n T_s + \phi[n]) + \eta[n]) \tag{2.7}$$

where $T_s = 1/f_s$ and the short notations $A[n] = A(nT_s)$, $\tau[n] = \tau(nT_s)$, $\phi[n] = \phi(nT_s)$, $y[n] = y(nT_s)$, and $\eta[n] = \eta(nT_s)$. Note, that after intermediate frequency conversion $\phi[n]$ also contains the clock instabilities of the local oscillator. $Q_b(x)$ denotes the quantization function which maps each value of x onto one of 2^b discrete values. Let us for the moment assume that b is infinite an therefore

$$Q_b(x) = x \tag{2.8}$$

In this case the n-th sampled signal of one tracking channel is given by

$$y[n] = A[n]\, c(nT_s - \tau[n])\, d(nT_s - \tau[n]) \cos(2\pi f_{\mathsf{IF}} n T_s + \phi[n]) + \eta[n]\,. \tag{2.9}$$

2.1.3 Baseband Conversion

For baseband conversion, we apply the same principle as for the intermediate frequency conversion. However, to keep the phase information we multiply the signal with sine and cosine such that we end with the complex baseband signal

$$y[n] = A[n]\, c(nT_s - \tau[n])\, d(nT_s - \tau[n])\, (\cos(\phi[n]) + j \sin(\phi[n])) \tag{2.10}$$
$$= A[n]\, e^{j\phi[n]} c(nT_s - \tau[n])\, d(nT_s - \tau[n]) + \eta[n] \tag{2.11}$$
$$= y_{\mathsf{I}}[n] + j y_{\mathsf{Q}}[n] \tag{2.12}$$

The bandwidth of the baseband signal (2.12) is smaller than the bandwidth of the intermediate frequency signal. Therefore, a decimation of the sampling rate f_s by the factor D can be applied, such that T_s becomes larger than for the IF signal.

2.2 Phase and Delay Model

In order to get a better understanding of the code delay and carrier phase and frequency we deeper discuss the phase and delay model in the following. The continuous time signal phase $\phi_{kl}(t)$ is influenced by different effects which allows to decompose

$$\phi_{kl}(t) = \frac{2\pi}{\lambda_{\mathsf{c},l}} \left(r_k(t) + c \left(\tau_{kl}^{\mathsf{clk,s}}(t) - \tau^{\mathsf{clk,r}}(t) \right) + \tau_{kl}^{\mathsf{tro}}(t) - \tau_{kl}^{\mathsf{ion}}(t) \right) + \eta_{kl}^{\phi} \tag{2.13}$$

where $\lambda_{\mathsf{c},l} = \frac{c}{f_{\mathsf{c},l}}$ and c is the speed of light. $r_k(t)$ denotes the distance from the k-th satellite to the receiver, i.e. the line-of-sight (LOS) distance, $\tau_{kl}^{\mathsf{clk,s}}(t)$, $\tau^{\mathsf{clk,r}}(t)$ denote the satellite and receiver clock offset, respectively, and $\tau_{kl}^{\mathsf{tro}}(t)$, $\tau_{kl}^{\mathsf{ion}}(t)$ denote the tropospheric and ionospheric delay, respectively. η_{kl}^{ϕ} is an additional higher order noise process. For the signal delay $\tau_{kl}(t)$ we can do a similar decomposition

$$\tau_{kl}(t) = \frac{1}{c} \left(r_k(t) + c \left(\tau_{kl}^{\mathsf{clk,s}}(t) - \tau_{kl}^{\mathsf{clk,r}}(t) \right) + \tau_{kl}^{\mathsf{tro}}(t) - \tau_{kl}^{\mathsf{ion}}(t) \right) + \eta_{kl}^{\phi} \tag{2.14}$$

To keep the notation simple, we will omit the subscripts $\bullet_k$ and $\bullet_l$ in the following and keep in mind that some effects are satellite or frequency dependent.

The models (2.13) and (2.14) can be use to derive a process model for $\phi(t)$ and $\tau(t)$ which can be used as apriori information in the parameter estimation process.

2.2.1 Satellite Movement and Doppler Model

In a first step we only consider the influence of the LOS distance $r_k(t)$ and on $\phi(t)$ and ignore all other terms in (2.13) which yields

$$\phi(t) = \frac{2\pi}{\lambda} r(t)\,, \tag{2.15}$$

where we omitted the subscript $\bullet_k$, $\bullet_l$ for the sake of readability. An algebraic expression for $r(t)$ is derived in Appendix A. For a static receiver and a satellite moving in a constant orbit around the earth the position, velocity, acceleration model illustrated in Figure 2.2 is well suited. In this case the following continuous time-invariant stochastic state space model for $r(t)$ holds

$$\underbrace{\begin{bmatrix} \frac{\partial r(t)}{\partial t} \\ \frac{\partial v_r(t)}{\partial t} \\ \frac{\partial a_r(t)}{\partial t} \end{bmatrix}}_{\dot{\boldsymbol{r}}(t)} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}}_{\boldsymbol{F}} \underbrace{\begin{bmatrix} r(t) \\ v_r(t) \\ a_r(t) \end{bmatrix}}_{\boldsymbol{r}(t)} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\boldsymbol{g}} \eta_a(t) \tag{2.16}$$

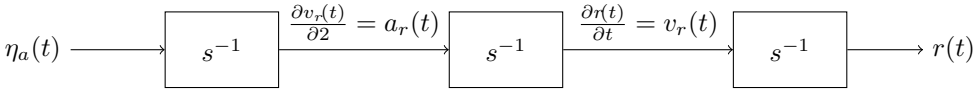


Figure 2.2: Continuous time LOS distance model

where $\eta_a(t)$ is the driving noise process for the LOS acceleration $a_r(t)$. The analytical solution to (2.16) is

$$\boldsymbol{r}(t) = e^{\boldsymbol{F}(t-t_0)}\boldsymbol{r}(t_0) + \int_{t_0}^{t} e^{\boldsymbol{F}(t-\tau)}\boldsymbol{g}\eta_a(t)\,\mathrm{d}\tau \tag{2.17}$$

where $e^{\boldsymbol{F}t}$ is the matrix exponential of $\boldsymbol{F}$ and $\boldsymbol{r}(t_0)$ is the initial condition. In order to derive the discrete time model

$$\boldsymbol{r}((k+1)T) = \boldsymbol{A}\boldsymbol{r}(kT) + \boldsymbol{\eta}(kT) \tag{2.18}$$

we insert the second order Taylor series expansion of $e^{\boldsymbol{F}t}$

$$e^{\boldsymbol{F}t} = \boldsymbol{I}_3 + \boldsymbol{F}t + \frac{\boldsymbol{F}^2}{2}t^2 \tag{2.19}$$

into (2.17) and use $t_0 = kT$ as starting point

$$\boldsymbol{r}[k+1] = \left(\boldsymbol{I}_3 + \boldsymbol{F}T + \frac{\boldsymbol{F}^2}{2}T^2\right)\boldsymbol{r}[k] + \int_{kT}^{(k+1)T}\left(\boldsymbol{I}_3 + \boldsymbol{F}\left((k+1)T-\tau\right) + \frac{\boldsymbol{F}^2}{2}\left((k+1)T-\tau\right)^2\right)\boldsymbol{g}\eta_a(t)\mathrm{d}\tau \tag{2.20}$$

$$= \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix}\boldsymbol{r}[k] + \int_{kT}^{(k+1)T}\begin{bmatrix} \frac{((k+1)T-\tau)^2}{2} \\ (k+1)T-\tau \\ 1 \end{bmatrix}\eta_a(t)\mathrm{d}\tau \tag{2.21}$$

$$= \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix}\boldsymbol{r}[k] + \boldsymbol{\eta}_a[k]\,. \tag{2.22}$$

$\boldsymbol{\eta}_a[k]$ denotes a third order noise process with covariance matrix $\boldsymbol{C}_{\boldsymbol{\eta}_a}$. The properties of this process are discussed in more detail in Section 2.2.2 and Appendix C Multiplying the left and right hand side of (2.22) with $\frac{2\pi}{\lambda} = \frac{2\pi f_c}{c}$ and substituting for

$$f_{\mathrm{D}}(t) = \frac{2\pi f_c}{c}v_r(t) \tag{2.23}$$

$$\alpha_{\mathrm{D}}(t) = \frac{2\pi f_c}{c}a_r(t) \tag{2.24}$$

yields the state space model known from literature [3-4 zitate]

$$\begin{bmatrix} \phi[k+1] \\ f_{\mathrm{D}}[k+1] \\ \alpha_{\mathrm{D}}[k+1] \end{bmatrix} = \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \phi[k] \\ f_{\mathrm{D}}[k] \\ \alpha_{\mathrm{D}}[k] \end{bmatrix} + \boldsymbol{\eta}_\alpha[k] \tag{2.25}$$

2.2.2 General State Space Model

Let $\phi(t) \in \mathbb{R}$ now be a sum of random walks of different order $\eta_i(t) \sim \mathcal{N}\left(0, \sigma_i^2\right)$ and temporal processes of different order $u_i(t) \in \mathbb{R}$. In this case it can generally be illustrated as given in Figure 2.3 and the following

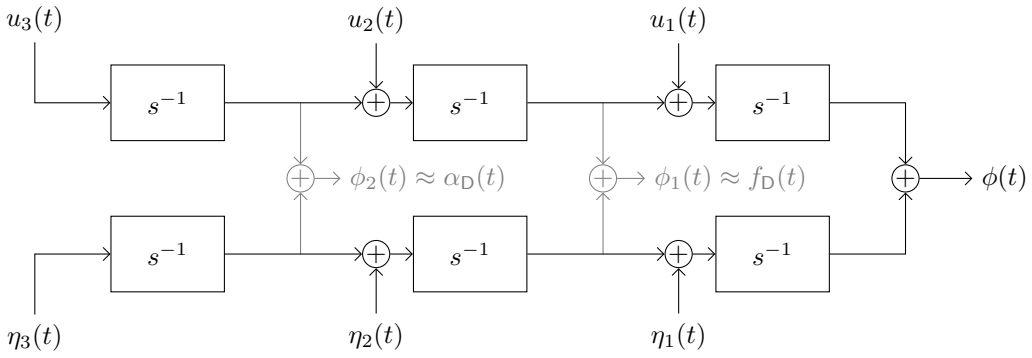


Figure 2.3: Illustration of the general state space model

continous time differential equation holds

$$\underbrace{\begin{bmatrix} \frac{\partial \phi_0(t)}{\partial t} \\ \frac{\partial \phi_1(t)}{\partial t} \\ \frac{\partial \phi_2(t)}{\partial t} \end{bmatrix}}_{\dot{\boldsymbol{\phi}}(t)} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}}_{\boldsymbol{F}} \underbrace{\begin{bmatrix} \phi_0(t) \\ \phi_1(t) \\ \phi_2(t) \end{bmatrix}}_{\boldsymbol{\phi}(t)} + \boldsymbol{u}(t) + \boldsymbol{\eta}(t) \tag{2.26}$$

where $\dot{\boldsymbol{\phi}}(t) = \frac{\partial}{\partial t}\boldsymbol{\phi}(t)$ and

$$\boldsymbol{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} \tag{2.27}$$

$$\boldsymbol{\eta}(t) = \begin{bmatrix} \eta_1(t) \\ \eta_2(t) \\ \eta_3(t) \end{bmatrix} \tag{2.28}$$

A solution to (2.26) with given start value $\phi(kT)$ can be derived as shown in Appendix B to

$$\boldsymbol{\phi}(t) = \boldsymbol{\phi}(kT)\, e^{\boldsymbol{A}(t-kT)} + \int_{kT}^{t} e^{\boldsymbol{A}(t-\tau)}\boldsymbol{u}(\tau)\mathrm{d}\tau + \int_{kT}^{t} e^{\boldsymbol{A}(t-\tau)}\boldsymbol{\eta}(\tau)\mathrm{d}\tau \tag{2.29}$$

Again we use the second order Taylor approximation (2.19) to approximate $e^{\boldsymbol{A}t}$ in (2.29) to get the solution for $t = (k+1)T$

$$\begin{aligned} \boldsymbol{\phi}[k+1] \approx & \left(\boldsymbol{I}_3 + \boldsymbol{F}T + \frac{\boldsymbol{F}^2}{2}T^2\right)\boldsymbol{\phi}[k] \\ & + \int_{kT}^{(k+1)T} \left(\boldsymbol{I}_3 + \boldsymbol{F}((k+1)T-\tau) + \frac{\boldsymbol{F}^2}{2}((k+1)T-\tau)^2\right)\boldsymbol{u}(\tau)\mathrm{d}\tau \\ & + \int_{kT}^{(k+1)T} \left(\boldsymbol{I}_3 + \boldsymbol{F}((k+1)T-\tau) + \frac{\boldsymbol{F}^2}{2}((k+1)T-\tau)^2\right)\boldsymbol{\eta}(\tau)\mathrm{d}\tau \end{aligned} \tag{2.30}$$

$$= \begin{bmatrix} 1 & T & T^2/2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix} \boldsymbol{\phi}[k] + \bar{\boldsymbol{u}}[k] + \bar{\boldsymbol{\eta}}[k] \tag{2.31}$$

where integrated noise and deterministic inputs are denoted

$$\bar{\boldsymbol{u}}[k] = \underbrace{\begin{bmatrix} \bar{u}_{1,1}[k] \\ 0 \\ 0 \end{bmatrix}}_{\bar{\boldsymbol{u}}_1[k]} + \underbrace{\begin{bmatrix} \bar{u}_{2,2}[k] \\ \bar{u}_{2,1}[k] \\ 0 \end{bmatrix}}_{\bar{\boldsymbol{u}}_1[k]} + \underbrace{\begin{bmatrix} \bar{u}_{3,3}[k] \\ \bar{u}_{3,2}[k] \\ \bar{u}_{3,1}[k] \end{bmatrix}}_{\bar{\boldsymbol{u}}_3[k]} \tag{2.32}$$

$$\bar{\boldsymbol{\eta}}[k] == \underbrace{\begin{bmatrix} \bar{\eta}_{1,1}[k] \\ 0 \\ 0 \end{bmatrix}}_{\bar{\boldsymbol{\eta}}_1[k]} + \underbrace{\begin{bmatrix} \bar{\eta}_{2,2}[k] \\ \bar{\eta}_{2,1}[k] \\ 0 \end{bmatrix}}_{\bar{\boldsymbol{\eta}}_2[k]} + \underbrace{\begin{bmatrix} \bar{\eta}_{3,3}[k] \\ \bar{\eta}_{3,2}[k] \\ \bar{\eta}_{3,1}[k] \end{bmatrix}}_{\bar{\boldsymbol{\eta}}_3[k]} . \tag{2.33}$$

In (2.32) and (2.33) we used the abbreviations

$$\bar{u}_{i,j}[k] = \frac{1}{(j-1)!} \int_{kT}^{(k+1)T} ((k+1)T - \tau)^{j-1} u_i(\tau) \mathrm{d}\tau = \frac{T^j}{j(j-1)} u_i[k] \tag{2.34}$$

$$\bar{\eta}_{i,j}[k] = \frac{1}{(j-1)!} \int_{kT}^{(k+1)T} ((k+1)T - \tau)^{j-1} \eta_i(\tau) \mathrm{d}\tau \tag{2.35}$$

where we assumed that $u_i(t)$ is constant in the interval $kT < t < (k+1)T$. The covariance matrix of $\bar{\boldsymbol{\eta}}[k]$ is derived in Appendix C as

$$\boldsymbol{C} = \sigma_1^2 \begin{bmatrix} T & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \sigma_2^2 \begin{bmatrix} \frac{T^3}{3} & \frac{T}{2} & 0 \\ \frac{T}{2} & T & 0 \\ 0 & 0 & 0 \end{bmatrix} + \sigma_3^2 \begin{bmatrix} \frac{T^5}{20} & \frac{T^4}{8} & \frac{T^3}{6} \\ \frac{T^4}{8} & \frac{T^3}{3} & \frac{T^2}{2} \\ \frac{T^3}{6} & \frac{T^2}{2} & T \end{bmatrix} \tag{2.36}$$

2.2.3 Receiver and Satellite Clock Model

2.2.4 Ionospheric Delay

2.2.5 Tropospheric Delay

Chapter 3

Parameter Estimation

3.1 Baseband Signal Processing in a Common GNSS Receiver

Figure 3.1 shows the general structure of the baseband signal processing unit of a GNSS receiver. The complex

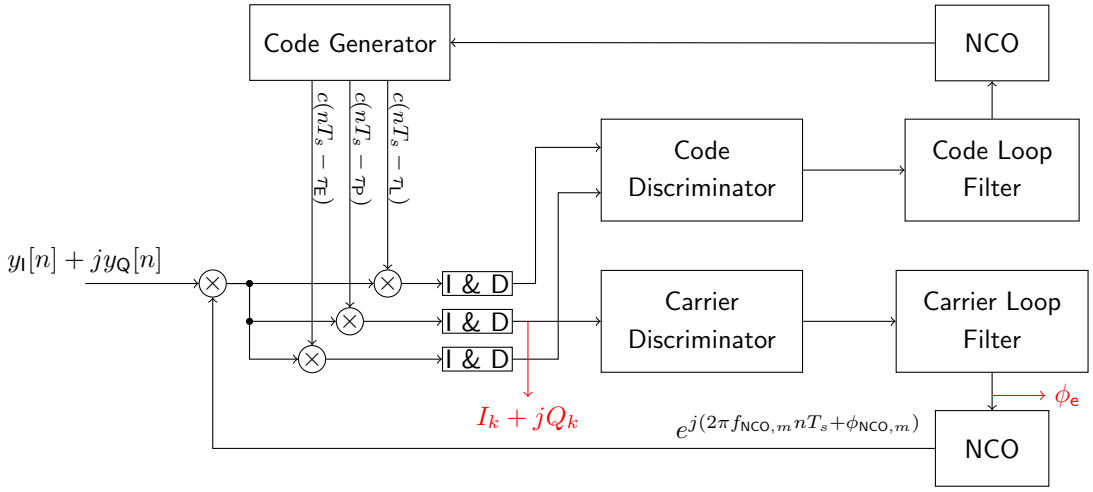


Figure 3.1: Illsutration of the Signal Processing Unit in the Case of Baseband Signals

baseband signal $y[n]$ is first multiplied with the complex phase $e^{j(2\pi f_{\text{NCO}}[m]nT_s+\phi_{\text{NCO}}[m])}$ where $f_{\text{NCO}}[m]$ and $\phi_{\text{NCO}}[m]$ are the numerically controlled oscillator (NCO) frequency and phase in the m-th integration interval, respectively

$$y'[n] = y[n]e^{-j(2\pi f_{\text{NCO}}[m]nT_s+\phi_{\text{NCO}}[m])} \tag{3.1}$$

$$= A[n]\, e^{j(\phi[n]-\phi_{\text{NCO}}[m]-2\pi f_{\text{NCO}}[m]nT_s)} c(nT_s-\tau[n])\, d(nT_s-\tau[n]) + \eta[n] \tag{3.2}$$

The complex output signal is than correlated with the early, prompt and late versions $c(nT_s-\tau_{\text{E}})$, $c(nT_s)$, $c(nT_s-\tau_{\text{L}})$ of the code replica. We use

$$\tau_{\text{P}} = \tau_{\text{NCO}} \tag{3.3}$$

$$\tau_{\text{E}} = \tau_{\text{NCO}} - \Delta_\tau \tag{3.4}$$

$$\tau_{\text{L}} = \tau_{\text{NCO}} + \Delta_\tau \tag{3.5}$$

The early, prompt and late correlator output signals are integrated over a time interval $T_{\mathsf{coh}} = MT_s$, such that M is the number of samples within the coherent integration period. The output of the prompt integrate and dump filter is given by

$$z[m] = \sum_{n=n_m+1}^{n_m+M} (y'[n]c(nT_s - \tau_{\mathsf{NCO}}[m])) \tag{3.6}$$

$$= \sum_{n=n_m+1}^{n_m+M} \left(A[n]\, e^{j(2\pi(f_{\mathsf{d}}[n]-f_{\mathsf{NCO}}[m])nT_s+\phi_0[n]-\phi_{\mathsf{NCO}}[m])} c(nT_s - \tau[n])\, d(nT_s - \tau[n]) + \eta[n] \right) c(nT_s - \tau_{\mathsf{NCO}}[m]) \tag{3.7}$$

$$\approx \frac{M\bar{A}[m]\, d[m]}{2} R_c(\tau_{\mathsf{e}}[m]) \left(\cos(2\pi f_{\mathsf{e}}[m]\, mT_{\mathsf{coh}} + \phi_{\mathsf{e}}[m]) + j \sin(2\pi f_{\mathsf{e}}[m]\, mT_{\mathsf{coh}} + \phi_{\mathsf{e}}[m])\right) + \eta_c[m] \tag{3.8}$$

where $\eta_c[m] \in \mathbb{C}$ denotes the noise after correlation which is zero mean correlated Gaussian noise with variance $\sigma^2_{\eta_c}$. The mean amplitude within the m-the integration interval is

$$\bar{A}[m] = \frac{1}{M} \sum_{n=n_m+1}^{n_m+M} A[n] \tag{3.9}$$

and $d[m] \in \{-1, 1\}$ is the symbol within this period, where we assume that we do not integrate over two different bit periods. The code auto-correlation function $R_c(\tau_1 - \tau_2)$ is defined

$$R_c(\tau_1 - \tau_2) = \sum_{n=m+1}^{m+M} c(nT_s - \tau_1)\, c(nT_s - \tau_2) \tag{3.10}$$

$\eta_c[m]$ can be separated into inphase and quadrature component

$$\eta_c[m] = \eta_{c,\mathsf{I}}[m] + j\eta_{c,\mathsf{Q}}[m] \tag{3.11}$$

with variance $\sigma^2_{\eta_{c,\mathsf{I}}} = \sigma^2_{\eta_{c,\mathsf{Q}}} = \frac{\sigma^2_{\eta_c}}{2}$. Using (3.8) and (3.11), the inphase and quadrature components are

$$I[m] = \frac{M\bar{A}[m]\, d[m]}{2} R_c(\tau_{\mathsf{e}}[m]) \cos(2\pi f_{\mathsf{e}}[m]\, mT_{\mathsf{coh}} + \phi_{\mathsf{e}}[m]) + \eta_{c,\mathsf{I}}[m] \tag{3.12}$$

$$Q[m] = \frac{M\bar{A}[m]\, d[m]}{2} R_c(\tau_{\mathsf{e}}[m]) \sin(2\pi f_{\mathsf{e}}[m]\, mT_{\mathsf{coh}} + \phi_{\mathsf{e}}[m]) + \eta_{c,\mathsf{Q}}[m] \tag{3.13}$$

We further introduce the error functions

$$\tau_{\mathsf{e}}[m] \approx \tau_{\mathsf{NCO}}[m] - \frac{1}{M} \sum_{n=n_m+1}^{n_m+M} \tau[n] \tag{3.14}$$

$$f_{\mathsf{e}}[m] \approx f_{\mathsf{NCO}}[m] - \frac{1}{M} \sum_{n=n_m+1}^{n_m+M} f_{\mathsf{d}}[n] \tag{3.15}$$

$$\phi_{\mathsf{e}}[m] \approx \phi_{\mathsf{NCO}}[m] - \frac{1}{M} \sum_{n=n_m+1}^{n_m+M} \phi_0[n] \tag{3.16}$$

is the error of the code delay estimate. The code and carrier discriminator deliver the code phase and carrier phase error estimates while code and carrier loop filters remove the noise from the phase error estimates. These error estimates are than forwarded to the respective NCOs. The NCO signals are than used for doppler removal or code generation which closes the loop.

Chapter 4

Calculation of Scintillation Parameters

4.1 Calculation of S_4

In general the scintillation intensity index S_4 is defined by

$$S_4 = \sqrt{\frac{\mathsf{E}\left(\rho^2[m]\right) - \mathsf{E}\left(\rho[m]\right)^2}{\mathsf{E}\left(\rho[m]\right)^2}} \tag{4.1}$$

where $\rho[m] = \bar{A}^2[m]$ is the squared signal amplitude or signal intensity. S_4 is the standard deviation of the signal intensity divided by its mean, i.e. the so-called coefficient of variation or relative standard deviation of the signal intensity. Let us for the moment assume perfect knowledge on code delay, and carrier frequency and phase, i.e. $\tau_\mathsf{e}[m] = 0$, $f_\mathsf{e}[m] = 0$, and $\phi_\mathsf{e}[m] = 0$. In this case (3.8) yields

$$I[m] = \frac{M\bar{A}[m]\,d[m]}{2} R_c(0) + \eta_{c,\mathsf{I}}[m] = \psi[m] + \eta_{c,\mathsf{I}}[m] \tag{4.2}$$

$$Q[m] = 0 + \eta_{c,\mathsf{Q}}[m] = \eta_{c,\mathsf{Q}}[m]\,. \tag{4.3}$$

where $Q[m]$ only contains noise, it is assumed that $\psi[m] = \frac{M\bar{A}[m]d[m]}{2} R_c(0)$ is constant over one bit period b and is denoted by $\psi[b]$. In the noise free case $I^2[m] \sim \rho[m]$ and $\psi^2[b] \sim \bar{A}^2[b]$, where $\forall_{m=1}^{M} \bar{A}^2[m] = \bar{A}^2[b]$. This can be used to extract a measure $\rho[b]$ for the signal intensity over the b-th bit period from M samples of $I[m]$. Let

$$P_I[b] = \left(\sum_{m=1}^{M} I[m]\right)^2 \tag{4.4}$$

$$\tilde{P}_I[b] = \sum_{m=1}^{M} I^2[m]\,, \tag{4.5}$$

be the wide and narrowband power $P_I[b]$ and $\tilde{P}_I[b]$ of the inphase component in the b-th bit period. M denotes the number of integrated correlator output samples within one bit period T_B. For example, for an integration time of $T_\mathsf{coh} = 1\,\mathrm{ms}$ and a bit period length of $T_\mathsf{B} = 20\,\mathrm{ms}$ we have $M = 20$. Inserting (4.2) into (4.4) and

(4.5) the wide band and narrow band power are given by

$$\begin{aligned} P_I[b] &= \left(\sum_{m=1}^{M} \left(\psi[b] + \eta_{c,\mathrm{I}}[m] \right) \right)^2 \\ &= \left(M\psi[b] + \sum_{m=1}^{M} \eta_{c,\mathrm{I}}[m] \right)^2 \\ &= M^2\psi^2[b] + 2M\psi[b] \sum_{m=1}^{M} \eta_{c,\mathrm{I}}[m] + \left(\sum_{m=1}^{M} \eta_{c,\mathrm{I}}[m] \right)^2 \\ &= M^2\psi^2[b] + 2M\psi[b] \sum_{m=1}^{M} \eta_{c,\mathrm{I}}[m] + \sum_{m=1}^{M} \eta_{c,\mathrm{I}}^2[m] + \sum_{\substack{p=1 \\ p\neq m}}^{M} \sum_{m=1}^{M} \eta_{c,\mathrm{I}}[m]\, \eta_{c,\mathrm{I}}[p] \end{aligned} \tag{4.6}$$

$$\begin{aligned} \tilde{P}_I[b] &= \sum_{m=1}^{M} \left(\psi[b] + \eta_{c,\mathrm{I}}[m] \right)^2 \\ &= \sum_{m=1}^{M} \left(\psi^2[b] + 2\psi[b]\, \eta_{c,\mathrm{I}}[m] + \eta_{c,\mathrm{I}}^2[m] \right) \\ &= M\psi^2[b] + 2\psi[b] \sum_{m=1}^{M} \eta_{c,\mathrm{I}}[m] + \sum_{m=1}^{M} \eta_{c,\mathrm{I}}^2[m] \end{aligned} \tag{4.7}$$

The difference between inphase narrow band and wide band power for the b-th bit period is

$$\rho_I[b] = P_I[b] - \tilde{P}_I[b] \tag{4.8}$$

$$= M\left(M-1\right)\psi^2[b] + 2\left(M-1\right)\psi[b] \sum_{m=1}^{M} \eta_{c,\mathrm{I}}[m] + \sum_{\substack{p=1 \\ p\neq m}}^{M} \sum_{m=1}^{M} \eta_{c,\mathrm{I}}[m]\, \eta_{c,\mathrm{I}}[p]\,. \tag{4.9}$$

which is in principle a good measure for the signal intensity of the b-th bit period as

$$\mathsf{E}(\rho_I[b]) = M\left(M-1\right)\psi^2[b] \tag{4.10}$$

is proportional to the squared signal amplitude $\bar{A}^2[b]$.

4.1.1 Elimination of Ambient Noise Influence

However, the expected value of $\rho_I^2[b]$ is not proportional to $\bar{A}^4[b]$ but includes an offset due to the noise variance. Therefore, inserting (4.9) directly into the S_4 formula (4.1) would lead to an increased estimate of S_4 due to the ambient noise. In order to calculate the S_4 due to ambient noise let us assume that $\forall_b \psi[b] = \psi$ for the

interval in which the expected values in (4.1) are calculated. In this case the expected value of $\rho_I^2[b]$ is

$$\begin{aligned}\mathsf{E}\left(\rho^2[b]\right) = \mathsf{E}\Bigg(& M^2\left(M-1\right)^2\psi^4 + 4\left(M-1\right)^2\left(\psi\sum_{m=1}^{M}\eta_{c,\mathsf{I}}[m]\right)^2 + \left(\sum_{\substack{p=1\\p\neq m}}^{M}\sum_{m=1}^{M}\eta_{c,\mathsf{I}}[m]\,\eta_{c,\mathsf{I}}[p]\right)^2 \\ & + 4M\left(M-1\right)^2\psi^3\sum_{m=1}^{M}\eta_{c,\mathsf{I}}[m] + 4\left(M-1\right)\psi\sum_{m=1}^{M}\eta_{c,\mathsf{I}}[m]\left(\sum_{\substack{p=1\\p\neq m}}^{M}\sum_{m=1}^{M}\eta_{c,\mathsf{I}}[m]\,\eta_{c,\mathsf{I}}[p]\right) \\ & + M\left(M-1\right)\psi^2\left(\sum_{\substack{p=1\\p\neq m}}^{M}\sum_{m=1}^{M}\eta_{c,\mathsf{I}}[m]\,\eta_{c,\mathsf{I}}[p]\right)\Bigg) \end{aligned} \tag{4.11}$$

$$\begin{aligned} = {} & M^2\left(M-1\right)^2\psi^4 + 4\left(M-1\right)^2\mathsf{E}\left(\psi^2\sum_{m=1}^{M}\eta_{c,\mathsf{I}}^2[m]\right) \\ & + \mathsf{E}\left(\sum_{\substack{p=1\\p\neq m}}^{M}\sum_{m=1}^{M}\left(\eta_{c,\mathsf{I}}[m]\,\eta_{c,\mathsf{I}}[p]\sum_{\substack{p=1\\p\neq m}}^{M}\sum_{m=1}^{M}\eta_{c,\mathsf{I}}[m]\,\eta_{c,\mathsf{I}}[p]\right)\right) \end{aligned} \tag{4.12}$$

$$= M^2\left(M-1\right)^2\psi^4 + 4M\left(M-1\right)^2\psi^2\frac{\sigma_n^2}{2} + 2M\left(M-1\right)\frac{\sigma_n^4}{2} \tag{4.13}$$

where it was used that

$$\mathsf{E}\left(\eta_{c,\mathsf{I}}^2[m]\right) = \frac{\sigma_n^2}{2}. \tag{4.14}$$

Inserting (4.13) and (**??**) into (4.1) we get the general expression for S_4 if the signal amplitude is constant, i.e. the ambient noise contribution to S_4 can be given as

$$S_{4,\mathsf{N}_0} = \sqrt{\frac{4M\left(M-1\right)^2\psi\frac{\sigma_n^2}{2} + 2M\left(M-1\right)\frac{\sigma_n^4}{2}}{M^2\left(M-1\right)^2\psi^4}} \tag{4.15}$$

$$= \sqrt{\frac{2\sigma_n^2}{M\psi^2} + \frac{\sigma_n^4}{M\left(M-1\right)\psi^4}} \tag{4.16}$$

$$= \sqrt{\frac{2}{M\frac{\psi^2}{\sigma_n^2}} + \frac{1}{M\left(M-1\right)\frac{\psi^4}{\sigma_n^4}}} \tag{4.17}$$

Note, that the carrier-to-noise ratio in the linear domain $\frac{\mathsf{C}}{\mathsf{N}_0}$ in the case of perfect code delay, and carrier frequency and phase knowledge is defined as [**?**]

$$\frac{\mathsf{C}}{\mathsf{N}_0} = \frac{A^2}{N_0} = \frac{\psi^2}{N_0} \tag{4.18}$$

and the noise power is given as

$$\sigma_n^2 = \frac{N_0}{T_{\mathsf{coh}}}. \tag{4.19}$$

Therefore, (4.17) can be rewritten

$$S_{4,\mathsf{N}_0} = \sqrt{\frac{2}{MT_{\mathsf{coh}}\frac{\mathsf{C}}{\mathsf{N}_0}}\left(1 + \frac{1}{2\left(M-1\right)T_{\mathsf{coh}}\frac{\mathsf{C}}{\mathsf{N}_0}}\right)}. \tag{4.20}$$

For a coherent integration time $T_{\text{coh}} = 1\,\text{ms}$ and $M = 20$ this leads to the well known S_4 due to ambient noise formula [**?**]

$$S_{4,\text{N}_0} = \sqrt{\frac{100}{\frac{\text{C}}{\text{N}_0}}\left(1 + \frac{500}{19\frac{\text{C}}{\text{N}_0}}\right)}. \tag{4.21}$$

The estimate for S_4 without noise dependency is therefore given by

$$S_4 = \sqrt{\frac{\text{E}\left(\rho_I^2[b]\right) - \text{E}\left(\rho_I[b]\right)^2}{\text{E}\left(\rho_I[b]\right)^2} - \frac{2}{MT_{\text{coh}}\frac{\text{C}}{\text{N}_0}}\left(1 + \frac{1}{2\left(M-1\right)T_{\text{coh}}\frac{\text{C}}{\text{N}_0}}\right)} \tag{4.22}$$

However, in the present of ionospheric scintillations, the code delay, and carrier frequency and phase errors $\tau_{\text{e}}[m]$, $f_{\text{e}}[m]$, and $\phi_{\text{e}}[m]$ can be significant. In this case the inphase and quadrature component are

$$I[m] = \frac{M\bar{A}[m]\,d[m]}{2} R_c(\tau_{\text{e}}[m]) \cos(2\pi f_{\text{e}}[m]\, mT_{\text{coh}} + \phi_{\text{e}}[m]) + \eta_{c,\text{I}}[m] \tag{4.23}$$

$$Q[m] = \frac{M\bar{A}[m]\,d[m]}{2} R_c(\tau_{\text{e}}[m]) \sin(2\pi f_{\text{e}}[m]\, mT_{\text{coh}} + \phi_{\text{e}}[m]) + \eta_{c,\text{Q}}[m]\,. \tag{4.24}$$

In this case the signal intensity is proportional to

$$\rho[m] \sim I^2[m] + Q^2[m]\,. \tag{4.25}$$

The narrow band and wide band power of the whole signal are in this case defined by

$$P[b] = \left(\sum_{m=1}^{M} I[m]\right)^2 + \left(\sum_{m=1}^{M} Q[m]\right)^2 \tag{4.26}$$

$$\tilde{P}[b] = \sum_{m=1}^{M} \left(I^2[m] + Q^2[m]\right), \tag{4.27}$$

and we use the difference

$$\rho[b] = P[b] - \tilde{P}[b] \tag{4.28}$$

$$\begin{aligned} &= M\left(M-1\right)\left(\psi^2[b] + \tilde{\psi}^2[b]\right) + 2\left(M-1\right)\left(\psi[b]\sum_{m=1}^{M}\eta_{c,\text{I}}[m] + \tilde{\psi}[b]\sum_{m=1}^{M}\eta_{c,\text{Q}}[m]\right) \\ &+ \sum_{\substack{p=1 \\ p\neq m}}^{M}\sum_{m=1}^{M}\left(\eta_{c,\text{I}}[m]\,\eta_{c,\text{I}}[p] + \eta_{c,\text{Q}}[m]\,\eta_{c,\text{Q}}[p]\right) \end{aligned} \tag{4.29}$$

to calculate the corrected S_4

$$S_4 = \sqrt{\frac{\text{E}\left(\rho^2[b]\right) - \text{E}\left(\rho[b]\right)^2}{\text{E}\left(\rho[b]\right)^2} - \frac{2}{MT_{\text{coh}}\frac{\text{C}}{\text{N}_0}}\left(1 + \frac{1}{2\left(M-1\right)T_{\text{coh}}\frac{\text{C}}{\text{N}_0}}\right)}. \tag{4.30}$$

4.1.2 Detrending the Data

The signal intensity $\rho[b]$ is not only varying due to scintillation effects, but for example also due to multipath or satellite and receiver movement. When measuring the S_4 therefore care should be taken such that there is no multipath and the receiver is static. However, the satellite movement is unavoidable therefore the data has to be detrended. In the literatur, different detrending methods have been studied [**?**]. In the software receiver

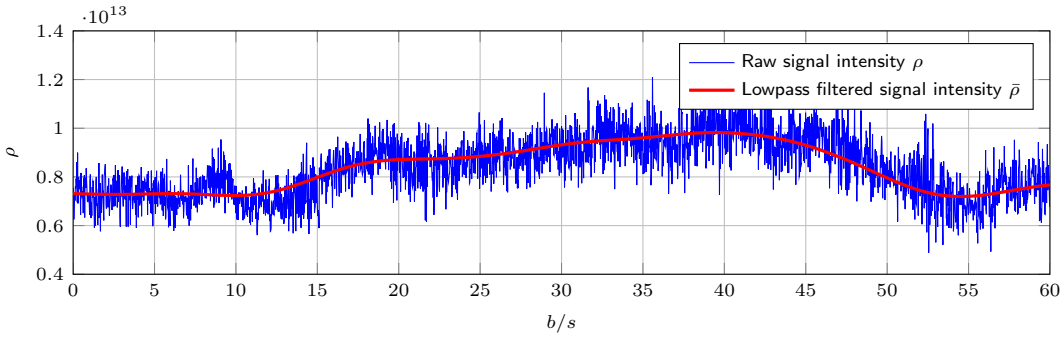


Figure 4.1: Raw and Lowpass Filtered signal intensity

we implemented the method where $\rho[b]$ is filtered with 6-th order Butterworth low pass filter $h[b]$ with cut-off frequency 0.1 Hz

$$\bar{\rho}[b] = h[b] * \rho[b] \tag{4.31}$$

and subsequently detrended

$$\rho_{\text{det}} = \frac{\rho[b]}{\bar{\rho}[b]}. \tag{4.32}$$

This detrended signal intensity estimate ρ_{det} is used in the S_4 formula (4.30). Figure 4.1 shows the raw and detrended signal intensity

4.2 Calculation of σ_ϕ

The scintillation phase variance is defined as

$$\sigma_\phi = \text{Var}\left(\phi\right), \tag{4.33}$$

where ϕ denotes the signal carrier phase. Let ϕ_{e} denotes the phase error output of the loop filter (see Figure 3.1). As for the calculation of S_4 the pase measurements have to be detrended in order to get rid of other effects such as satellite movement or inaccurate clocks. The carrier phase measurement of the i-th tracking channel can be modeled as

$$\phi^2_{\text{e},i} = \phi_{\text{sat},i} + \phi_{\text{clc}} + \phi_{\text{sci},i} + \eta_i \tag{4.34}$$

where $\phi_{\text{sat},i}$ denotes the phase error due to satellite movement, ϕ_{clc} denotes the phase error due to clock instability and $\phi_{\text{sci},i}$ denotes the phase error due to scintillations. $\eta_{\phi,i}$ is white Gaussian noise with variance $\sigma^2_{\eta_{\phi,i}}$.

4.2.1 Clock Error Removal

In general a stable clock which leads to a small error ϕ_{clc} should be used. However, in case this is not available or a more accurate estimate is desired the error can be compensated if there is one tracking channel, i.e. a signal from one satellite which does not exhibit phase scintillations. In this case the difference [**?**]

$$\phi_{\text{e},i\text{-ref}} = \phi_{\text{e},i} - \phi_{\text{e,ref}} \tag{4.35}$$

$$= \phi_{\text{sat},i} - \phi_{\text{sat,ref}} + \phi_{\text{sci},i} + \eta_{\phi,i} - \eta_{\phi,\text{ref}} \tag{4.36}$$

where $\bullet_{\text{ref}}$ denotes the reference channel can be used for σ_ϕ estimation.

4.2.2 Geometry Error Removal

The error due to satellite movement can be modeled as a third order polynomial for a static receiver. The same holds for the difference $\phi_{\text{sat},i} - \phi_{\text{sat,ref}}$. Therefore this term can be eliminated by subtracting a third order polynomial fit $\phi^{\text{3rd}}_{\text{e},i\text{-ref}}$ to $\phi_{\text{e},i\text{-ref}}$ from $\phi_{\text{e},i\text{-ref}}$ to obtain

$$\phi_{\text{det},i} \approx \phi_{\text{sci},i} + \eta_{\phi,i} - \eta_{\phi,\text{ref}} \tag{4.37}$$

4.2.3 Low Frequency Component Removal

As in the case of the S_4 estimation the high pass filtering is applied in order to remove all remaining low frequency components from the phase, as these are commonly not due to scintillation. For high pass filtering a cascade of three second order butterworth highpass filters $h_{\text{high},2}(t)$, i.e. a sixth order butterworth high pass filter is applied. This leads to the ϕ values of the n-th tracking channel

$$\phi_{=} h_{\text{high},2}(t) * h_{\text{high},2}(t) * h_{\text{high},2}(t) * \phi_{\text{det},i} \tag{4.38}$$

Bibliography

Appendices

Appendix A

Satellite Movement and Doppler Model

Figure A.1 shows the movement of a satellite S_k on its orbit around the earth. v_S denotes the constant satellite's tangential velocity, while $v_R(t)$ denotes the velocity of the satellite relative to the receiver

$$v_R(t) = \cos(\beta(t))\, v_S, \tag{A.1}$$

i.e. the LOS velocity. Further, $d(t)$, r_S, and r_e denote the distance between satellite and receiver, the satellite height and the earth radius, respectively. Due to the relative velocity between satellite and receiver v_R, which is different from zero if the satellite is not in the zenith, the distance between satellite and receiver varies between transmission and reception time of the receiver. This leads to the doppler effect, i.e. the received frequency is shifted by a time dependent doppler offset $f_{D,l}(t)$ which is defined

$$f_{D,l}(t) = \frac{v_R(t)}{c} f_{c,l}. \tag{A.2}$$

In order to calculate v_R we first note that the time-varying distance between satellite and receiver $r_k(t)$ in dependency of α, r_e and r_S is

$$r_k(t) = \sqrt{r_e^2 + r_S^2 + r_e r_S \cos(\beta(t))}. \tag{A.3}$$

The the connection between sine and cosine, the relations shown in Figure A.1, and the sine law further yield

$$\cos\beta(t) = \sin(90 - \beta(t)) = \frac{r_e \sin\alpha(t)}{r_k(t)} \tag{A.4}$$

and therefore by inserting (A.4) into (A.1)

$$v_R(t) = \frac{r_e \sin\alpha(t)}{\sqrt{r_e^2 + r_S^2 + r_e r_S \cos(\alpha(t))}} v_S. \tag{A.5}$$

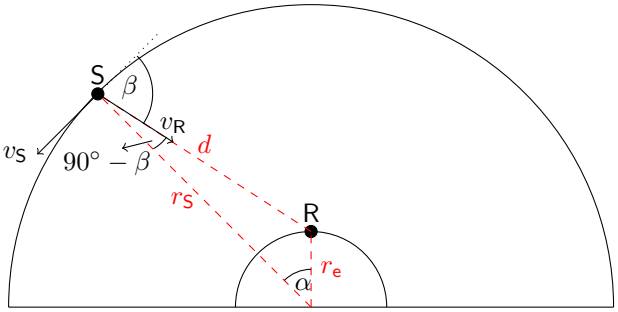


Figure A.1: Satellite Movement around the Earth

Inserting (A.5) into (A.6) yields the analytic expression for the doppler shift

$$f_{\mathrm{D,l}}(t) = v_{\mathsf{S}} \frac{r_{\mathsf{e}} \sin \alpha(t)}{\sqrt{r_{\mathsf{e}}^2 + r_{\mathsf{S}}^2 + r_{\mathsf{e}} r_{\mathsf{S}} \cos(\alpha(t))}} \frac{f_{\mathsf{c},l}}{c}. \tag{A.6}$$

Appendix B

Solution of the Continuous Time State Space Equation

In the following we shortly review the general solution to a differential equation of the form

$$\dot{\boldsymbol{y}}(t) = \boldsymbol{A}\boldsymbol{y}(t) + \boldsymbol{b}(t) \tag{B.1}$$

with a known start value $\boldsymbol{y}(kT)$ at $t = kT$. In this case we can manipulate (B.1) as follows

$$\dot{\boldsymbol{y}}(t) = \boldsymbol{A}\boldsymbol{y}(t) + \boldsymbol{b}(t) \quad \Big| \cdot e^{\int -\boldsymbol{A}\mathrm{d}t} \tag{B.2}$$

$$\dot{\boldsymbol{y}}(t)\, e^{\int -\boldsymbol{A}\mathrm{d}t} = \boldsymbol{A}\boldsymbol{y}(t)\, e^{\int -\boldsymbol{A}\mathrm{d}t} + \boldsymbol{b}(t)\, e^{\int -\boldsymbol{A}\mathrm{d}t} \quad \Big| -\boldsymbol{A}\boldsymbol{y}(t)\, e^{\int -\boldsymbol{A}\mathrm{d}t} \tag{B.3}$$

$$\dot{\boldsymbol{y}}(t)\, e^{\int -\boldsymbol{A}\mathrm{d}t} - \boldsymbol{A}\boldsymbol{y}(t)\, e^{\int -\boldsymbol{A}\mathrm{d}t} = \boldsymbol{b}(t)\, e^{\int -\boldsymbol{A}\mathrm{d}t} \tag{B.4}$$

$$\frac{\partial}{\partial t}\left(\boldsymbol{y}(t)\, e^{\int -\boldsymbol{A}\mathrm{d}t}\right) = \boldsymbol{b}(t)\, e^{\int -\boldsymbol{A}\mathrm{d}t} \quad \Big|_{kT}^{T} \tag{B.5}$$

$$\int_{kT}^{t} \frac{\partial}{\partial \tau}\boldsymbol{y}(\tau)\, e^{-\boldsymbol{A}\tau}\mathrm{d}\tau = \int_{kT}^{t} \boldsymbol{b}(\tau)\, e^{-\boldsymbol{A}\tau}\mathrm{d}\tau \tag{B.6}$$

$$\boldsymbol{y}(t)\, e^{-\boldsymbol{A}t} - \boldsymbol{y}(kT)\, e^{-\boldsymbol{A}kT} = \int_{kT}^{t} \boldsymbol{b}(\tau)\, e^{-\boldsymbol{A}\tau}\mathrm{d}\tau \tag{B.7}$$

$$\boldsymbol{y}(t) = \boldsymbol{y}(kT)\, e^{\boldsymbol{A}(t-kT)} + \int_{kT}^{t} \boldsymbol{b}(\tau)\, e^{\boldsymbol{A}(t-\tau)}\mathrm{d}\tau \tag{B.8}$$

Appendix C

Derivation of the Noise Statistics

Let the $\eta_i(t)$ be zero mean white Gaussian process with variances $\sigma^2_{\eta_i}$. Further we assume that the respective noise processes are uncorrelated, i.e.

$$\mathsf{E}[\eta_{i_1}(t)\,\eta_{i_2}(t)] = \delta_{i_1,i_2} = \begin{cases} 1 & i_1 = i_2 \\ 0 & \text{else} \end{cases} \tag{C.1}$$

The expected value $\mathsf{E}[\bar{\eta}_{i,j}(kT)]$ is zero as $\eta_i(t)$ is zero mean and

$$\mathsf{E}[\bar{\eta}_{i,j}(k_1T)\,\bar{\eta}_{i,j}(k_2T)] = 0 \tag{C.2}$$

Therefore, the covariance of the noise contribution can be calculated

$$\begin{aligned}
&\mathsf{E}[\bar{u}_{i_1,j_1}(k_1T)\,\bar{u}_{i_2,j_2}(k_1T)] \\
&= \mathsf{E}\left[\frac{1}{(j_1-1)!}\int_{kT}^{(k+1)T}((k+1)T-\tau)^{j_1-1}\eta_{i_1}(\tau)\mathrm{d}\tau\frac{1}{(j_2-1)!}\int_{kT}^{(k+1)T}((k+1)T-\tau)^{j_2-1}\eta_{i_2}(\tau)\mathrm{d}\tau\right] && \text{(C.3)} \\
&= \mathsf{E}\left[\frac{1}{(j_1-1)!(j_2-1)!}\int_{kT}^{(k+1)T}\int_{kT}^{(k+1)T}((k+1)T-\tau')^{j_1-1}\eta_{i_1}(\tau')\,((k+1)T-\tau)^{j_2-1}\eta_{i_2}(\tau)\mathrm{d}\tau'\mathrm{d}\tau\right] && \text{(C.4)} \\
&= \frac{1}{(j_1-1)!(j_2-1)!}\int_{kT}^{(k+1)T}\int_{kT}^{(k+1)T}((k+1)T-\tau')^{j_1-1}((k+1)T-\tau)^{j_2-1}\mathsf{E}[\eta_{i_1}(\tau')\,\eta_{i_2}(\tau)]\mathrm{d}\tau'\mathrm{d}\tau && \text{(C.5)} \\
&= \sigma^2_{\eta_{i_1}}\frac{1}{(j_1-1)!\,(j_2-1)!}\delta_{i_1,i_2}\int_{kT}^{(k+1)T}\int_{kT}^{(k+1)T}((k+1)T-s)^{j_1+j_2-2}\,\mathrm{d}s\mathrm{d}s && \text{(C.6)} \\
&= \sigma^2_{\eta_{i_1}}\frac{T^{j_1+j_2-1}}{(j_1-1)!\,(j_2-1)!\,(j_1+j_2-1)}\delta_{i_1,i_2} && \text{(C.7)}
\end{aligned}$$

Therefore the covariance matrix of the noise can be expressed by

$$\boldsymbol{Q} = \mathsf{E}\left[\sum_{i=1}^{3}\bar{\boldsymbol{\eta}}_i\sum_{i=1}^{3}\bar{\boldsymbol{\eta}}_i^{\mathsf{H}}\right] = \sum_{i=1}^{3}\mathsf{E}\left[\bar{\boldsymbol{\eta}}_i\bar{\boldsymbol{\eta}}_i^{\mathsf{H}}\right] \tag{C.8}$$

$$= \sigma_1^2\begin{bmatrix} T & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \sigma_2^2\begin{bmatrix} \frac{T^3}{3} & \frac{T}{2} & 0 \\ \frac{T}{2} & T & 0 \\ 0 & 0 & 0 \end{bmatrix} + \sigma_3^2\begin{bmatrix} \frac{T^5}{20} & \frac{T^4}{8} & \frac{T^3}{6} \\ \frac{T^4}{8} & \frac{T^3}{3} & \frac{T^2}{2} \\ \frac{T^3}{6} & \frac{T^2}{2} & T \end{bmatrix} \tag{C.9}$$